Supplementary Material Toward a Reproducible GIS Quality Assurance Framework: Quantifying Error Mechanisms Across the Spatial Workflow Through Ten Illustrative Cases

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Supplementary Table S1. Traceability of the cases used in the Sobol sensitivity analysis presented in Figure 8.

Case No.	Reference	Distinct Scenario or Analytical Case	GIS Error / Uncertainty Category	Case Classification	Spatial Data or GIS Operation	Input Factor Relevant to Figure 8	Output / Response Variable	Metric or Analytical Method	Role in Sobol Analysis	Evidence Supporting the Case	Notes
1	Aller et al. (1987)	DRASTIC parameter weight subjectivity in groundwater vulnerability assessment	Modeling	Qualitative and illustrative	DRASTIC groundwater vulnerability model; seven-parameter weighted rating index	Weights	Groundwater vulnerability index sensitivity to expert-assigned parameter weights	DRASTIC rating index = sum(parameter rating * weight); weight range 1-5	Defines the DRASTIC weight parameters (values 1-5 per parameter) that are the direct source of the 'Weights' factor tested in the Sobol analysis (Figure 8; S1=0.30, ST=0.40)	Aller et al. (1987), EPA/600/2-87/035, Chapter 3: DRASTIC parameter weight assignments and rating descriptions	Primary source for DRASTIC model structure; subjective weight assignment is the 'Weights' Sobol factor
2	Aller et al. (1987)	Data availability constraints on DRASTIC input layers	Data Quality	Qualitative	Hydrogeological data layers; seven DRASTIC input parameters	Data availability	Model completeness as a function of available hydrogeological data	Parameter coverage; data source requirements for each DRASTIC input layer	Illustrates that data gaps constrain the 'Data availability' Sobol factor (Figure 8; S1=0.40, ST=0.50) -- the highest-sensitivity factor in the analysis	Aller et al. (1987), EPA/600/2-87/035, Chapter 2: DRASTIC data requirements and recommended data sources	Data availability constraints are directly modelled as the dominant Sobol factor (S1=0.40)
3	Aller et al. (1987)	Topography (T) parameter scale dependency: DEM resolution affects slope classification	Scale/Resolution	Quantitative and illustrative	Digital elevation model; DRASTIC topography parameter (slope gradient)	Scale	DRASTIC T-parameter rating class as a function of DEM spatial resolution	Rating class assignment; slope gradient percentage derived from DEM at varying resolutions	Demonstrates that DEM resolution affects the T-parameter rating class, contributing to the 'Scale' Sobol factor (Figure 8; S1=0.20, ST=0.25)	Aller et al. (1987), EPA/600/2-87/035, Table B-7: topography (T) parameter rating criteria by slope class	Scale dependency of DEM-derived topography directly illustrates the Scale factor in Figure 8
4	Heuvelink (1998)	Analytical error propagation through raster overlay operations	Data Quality	Quantitative	Raster overlay; Boolean and arithmetic map operations on spatial layers	Uncertainty	Output error variance as a function of input layer positional and attribute errors	Error propagation formula; output variance from combined input error variances	Provides the theoretical basis for quantifying uncertainty accumulation through GIS overlay, informing the 'Uncertainty' Sobol factor (Figure 8; S1=0.10, ST=0.15)	Heuvelink (1998), Taylor and Francis: Chapter 4, analytical error propagation for GIS operations	Foundational uncertainty propagation reference; Uncertainty factor S1=0.10 in Figure 8

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5	Heuvelink (1998)	Monte Carlo stochastic simulation of positional error in GIS spatial data	Data Quality	Quantitative	Vector spatial layers; stochastic positional error simulation	Uncertainty	Distribution of output values under stochastic positional error; mean, variance, and confidence interval	Monte Carlo simulation; mean and variance of output distribution across realisations	The Monte Carlo stochastic simulation approach underpins the N=10,000 Saltelli sampling strategy used in the Sobol analysis (Figure 8)	Heuvelink (1998), Taylor and Francis: Chapter 5, stochastic simulation of spatial data errors	Stochastic simulation methodology is directly analogous to Saltelli sampling in Figure 8
6	Heuvelink (1998)	Error propagation in buffer and proximity analysis operations	Data Quality	Quantitative	Buffer zones; proximity analysis; vector boundary operations	Scale	Boundary uncertainty width as a function of positional error magnitude and spatial scale	Error band width; buffer boundary uncertainty at varying scales	Scale-dependent boundary uncertainty in buffer operations informs the 'Scale' Sobol factor parameterisation (Figure 8; S1=0.20)	Heuvelink (1998), Taylor and Francis: Chapter 6, error in spatial analysis operations including buffering	Buffer boundary uncertainty illustrates scale-dependent error contributing to Scale factor in Figure 8
7	Heuvelink (1998)	Stochastic error simulation for attribute uncertainty in land-cover classification	Attributes	Quantitative	Land-cover classification; categorical attribute data quality	Uncertainty	Classification accuracy under varying attribute uncertainty scenarios	Confusion matrix; overall accuracy; variance of class proportions across realisations	Attribute uncertainty framework informs the 'Uncertainty' Sobol factor in Figure 8; attribute errors are one of the eight GIS error categories in the pairwise interdependency analysis	Heuvelink (1998), Taylor and Francis: Chapter 7, attribute error in categorical spatial data	Attribute error propagation contributes to the Uncertainty factor parameterisation (S1=0.10)
8	Crosetto and Tarantola (2001)	Variance-based sensitivity analysis of GIS environmental model parameters	Modeling	Quantitative	Environmental GIS suitability model; multi-parameter sensitivity analysis	Weights	Sobol first-order index S1 and total-effect index ST for GIS model input parameters	Sobol S1 and ST indices; variance decomposition by input factor	Direct methodological precedent for the variance-based Sobol sensitivity analysis applied in Figure 8; establishes the framework for decomposing model output variance by input factor	Crosetto and Tarantola (2001), International Journal of Geographical Information Science 15(5): 415-437; Sections 3-4 on variance-based sensitivity decomposition	This paper is the primary GIS-specific precedent for the Sobol approach used in Figure 8
9	Crosetto and Tarantola (2001)	Uncertainty propagation from input factor interactions in GIS environmental models	Modeling	Quantitative	Multi-factor GIS environmental model; input factor interaction effects	Uncertainty	Proportion of output variance attributable to factor interactions (ST minus S1)	Sobol total-effect index ST; interaction-effect proportion = ST - S1	The ST-S1 interaction terms in Figure 8 (e.g., Data availability: 0.50-0.40=0.10) reflect interaction effects quantified using the framework documented in this paper	Crosetto and Tarantola (2001), International Journal of Geographical Information Science 15(5): 415-437; Section 3.2 on higher-order interaction indices	ST minus S1 interaction analysis is explicitly documented for GIS models in this reference

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10	Fotheringham and Wong (1991)	MAUP aggregation effect on multivariate spatial statistics	Scale/Resolution	Quantitative	Census data; areal unit aggregation; multivariate regression analysis	Scale	Instability of regression coefficients under varying spatial aggregation levels	Regression coefficient variance across aggregation scales; Pearson r	Demonstrates that aggregation scale directly affects spatial model outputs, providing empirical support for the 'Scale' Sobol factor (Figure 8; S1=0.20, ST=0.25)	Fotheringham and Wong (1991), Environment and Planning A 23(7): 1025-1044; Tables 2-4 on coefficient instability across scales	MAUP scale effect provides empirical basis for Scale factor sensitivity in Figure 8
11	Fotheringham and Wong (1991)	MAUP zonation effect on spatial pattern detection	Scale/Resolution	Quantitative	Census zone boundaries; alternative spatial aggregation schemes; spatial autocorrelation	Weights	Variation in Moran I statistic across alternative zone boundary configurations	Moran I variance; spatial autocorrelation coefficient across zonation schemes	Zonation effects on pattern detection are analogous to boundary weight assignments in DRASTIC, linking the MAUP zonation problem to the 'Weights' Sobol factor (Figure 8; S1=0.30)	Fotheringham and Wong (1991), Environment and Planning A 23(7): 1025-1044; Section 4 on zonation effects	Boundary weight assignment variation is a direct analogue to the Weights factor in the Sobol analysis
12	Openshaw (1984)	MAUP scale problem: correlation coefficient varies with spatial aggregation level	Scale/Resolution	Quantitative	Areal spatial data; multiple spatial aggregation levels	Scale	Pearson correlation coefficient as a function of spatial aggregation scale	Pearson r across aggregation levels; scale-dependent coefficient variability	Foundational MAUP scale problem: demonstrates that scale is a primary GIS error source and directly motivates Scale as a Sobol sensitivity factor (Figure 8; S1=0.20)	Openshaw (1984), Geo Books, The Modifiable Areal Unit Problem: Sections 2-3 on the scale problem	Classic reference establishing scale as a fundamental GIS error category; supports S1(Scale)=0.20
13	Openshaw (1984)	MAUP zoning problem: alternative boundary configurations produce wide outcome ranges	Scale/Resolution	Quantitative	Areal spatial units; alternative boundary delineation configurations	Weights	Range of outcome variable values across alternative zoning configurations at fixed scale	Outcome variable variance across zonation schemes; sensitivity to boundary specification	Zone boundary specifications are a form of weight assignment; the zoning problem directly parallels the 'Weights' Sobol factor (Figure 8; S1=0.30, ST=0.40)	Openshaw (1984), Geo Books, The Modifiable Areal Unit Problem: Sections 4-5 on the zoning problem	Zoning problem underpins the Weights factor as a primary sensitivity parameter in Figure 8
14	O'Connor et al. (2023)	Multiscale GWR and MAUP mitigation: residual scale sensitivity persists	Scale/Resolution	Quantitative	Geographically weighted regression; spatial regression; bandwidth selection	Scale	AIC and residual standard error under varying bandwidth scales in multiscale GWR	AIC; residual SE; bandwidth sensitivity analysis in multiscale GWR	Documents that scale sensitivity persists in spatial models even after MAUP-mitigation methods are applied, reinforcing the inclusion of Scale as a Sobol factor (Figure 8; S1=0.20)	O'Connor et al. (2023), Geographical Analysis 55(3): 397-419; Results and Discussion sections	Contemporary evidence for persistent scale sensitivity supports Scale factor role in Figure 8

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15	Snyder (1987)	Conformal projection area distortion: Mercator scale factor k = sec(phi)	CRS/Datum	Quantitative	Map projections; Mercator conformal projection; area distortion by latitude	Scale	Percentage area error as a function of latitude for conformal map projections	Scale factor k = sec(phi); percentage area distortion formula	Projection-induced area distortion is a component of the 'Scale' Sobol factor; CRS choice affects area calculations used as DRASTIC model inputs (Figure 8; S1=0.20)	Snyder (1987), USGS Professional Paper 1395: Chapter 7, Mercator projection; distortion formulas	Projection scale distortion contributes to Scale factor in the Sobol analysis
16	Snyder (1987)	Equal-area projection accuracy for regional GIS analysis	CRS/Datum	Quantitative	Equal-area map projections; Albers conic projection; area calculation accuracy	Data availability	Percentage area error vs. ellipsoidal reference for equal-area projections	Area error percentage relative to ellipsoid reference; projection accuracy assessment	Projection choice affects the accuracy of spatial extent calculations used in DRASTIC; accurate area data is a component of the 'Data availability' Sobol factor (Figure 8; S1=0.40)	Snyder (1987), USGS Professional Paper 1395: Chapter 20, Albers equal-area conic; accuracy section	Equal-area projection accuracy informs Data availability quality of DRASTIC input layers
17	Usery et al. (2002)	Raster cell size effects on polygon area calculations	Scale/Resolution	Quantitative	Raster spatial data; cell size; polygon area computation in raster GIS	Scale	Percentage area error relative to vector reference as a function of raster cell size	Percentage area error vs. vector polygon area; resolution-dependent bias	Quantifies Scale factor impact on raster-derived area calculations (e.g., aquifer extent in DRASTIC); supports S1(Scale)=0.20 in Figure 8	Usery et al. (2002), Cartography and Geographic Information Science 29(3): 191-201; Tables 2-3	Provides quantitative evidence for raster scale effects on area-based DRASTIC parameters
18	Usery et al. (2002)	Multi-resolution data integration errors: resolution mismatch between input layers	Data Quality	Quantitative	Multi-source raster data integration; resolution mismatch between co-registered layers	Data availability	Positional discrepancy between co-registered layers of different spatial resolutions	Positional discrepancy magnitude; integration error across resolution combinations	Resolution mismatch reduces the effective data availability quality of DRASTIC input layers, contributing to the 'Data availability' Sobol factor (Figure 8; S1=0.40)	Usery et al. (2002), Cartography and Geographic Information Science 29(3): 191-201; Section 3 on data integration	Multi-source integration errors are a data availability concern for DRASTIC parameterisation
19	Li and Heap (2014)	Kriging interpolation uncertainty: kriging variance as spatial uncertainty measure	Modeling	Quantitative	Point spatial data; kriging spatial interpolation; variogram modelling	Uncertainty	Kriging variance distribution as a spatial uncertainty estimate at interpolated locations	Root mean squared prediction error (RMSPE); kriging variance spatial distribution	Interpolation uncertainty is a component of the 'Uncertainty' Sobol factor; kriging variance quantifies the residual uncertainty in spatially continuous DRASTIC inputs (Figure 8; S1=0.10)	Li and Heap (2014), Environmental Modelling and Software 53: 173-189; Section 2.3 on kriging uncertainty	Kriging variance provides spatial uncertainty estimates for interpolated DRASTIC input layers

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20	Li and Heap (2014)	IDW interpolation sensitivity to power parameter p	Modeling	Quantitative	Point spatial data; inverse distance weighting spatial interpolation	Weights	RMSE as a function of IDW power parameter p across cross-validation folds	RMSE; leave-one-out cross-validation across p values	The IDW power parameter is a weighting factor directly analogous to the DRASTIC weight parameters; sensitivity of outputs to p supports the 'Weights' Sobol factor (Figure 8; S1=0.30)	Li and Heap (2014), Environmental Modelling and Software 53: 173-189; Section 3.1 on IDW parameter sensitivity	IDW power parameter sensitivity is analogous to DRASTIC weight sensitivity (Weights factor S1=0.30)
21	Li and Heap (2014)	Cross-validation framework for spatial interpolation model comparison	Modeling	Quantitative	Spatial interpolation methods; cross-validation; comparative performance assessment	Uncertainty	RMSE and MAE for multiple interpolation methods under cross-validation	RMSE; MAE; leave-one-out cross-validation split across methods	Cross-validation uncertainty quantification informs the 'Uncertainty' Sobol factor parameterisation; provides a framework for comparing interpolation-induced errors in DRASTIC inputs (Figure 8)	Li and Heap (2014), Environmental Modelling and Software 53: 173-189; Section 2.4 on cross-validation methodology	Cross-validation framework supports uncertainty quantification for the Uncertainty Sobol factor
22	Shi (2008)	Positional uncertainty modelling using the epsilon-band concept	Data Quality	Quantitative	Vector spatial data; positional error; probabilistic epsilon-band model	Uncertainty	Error band width as a function of positional uncertainty parameters	Error band width; probabilistic positional uncertainty model parameters	Provides a formal model for positional uncertainty that informs the 'Uncertainty' Sobol factor parameterisation in Figure 8 (S1=0.10, ST=0.15)	Shi (2008), CRC Press, ISBN 978-1-4200-5927-4: Chapter 3, positional uncertainty modelling	Epsilon-band model provides the formal basis for positional uncertainty characterisation
23	Shi (2008)	Attribute uncertainty propagation through spatial operations	Attributes	Quantitative	Attribute data; spatial overlay operations; attribute error propagation	Uncertainty	Variance of derived attribute values after spatial operations under attribute uncertainty	Error variance of derived attributes; analytical propagation formula for attribute errors	Attribute uncertainty propagation contributes to the 'Uncertainty' Sobol factor; attribute errors are a documented error category in the pairwise interdependency matrix underlying Figure 8	Shi (2008), CRC Press, ISBN 978-1-4200-5927-4: Chapter 5, attribute uncertainty in GIS operations	Attribute error propagation is a component of the Uncertainty Sobol factor (S1=0.10) in Figure 8
24	ISO 19157:2013	Positional accuracy as a standardised data quality element: RMSE and CE90	Data Quality	Quantitative	Vector spatial data; positional accuracy	Data availability	RMSE and CE90 as standardised positional	RMSE; CE90 (circular error at 90% probability);	Standardised data quality metrics define the measurement	ISO 19157:2013, Section 4.14 (positional accuracy); Section 7	ISO 19157 provides the formal data quality framework referenced

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					assessment; coordinate accuracy measurement		accuracy metrics for spatial data quality	ISO data quality measures	framework for the 'Data availability' Sobol factor; data meeting positional accuracy thresholds is available for DRASTIC parameterisation (Figure 8; S1=0.40)	(data quality measures and evaluation procedures)	for DRASTIC data availability assessment
25	ISO 19157:2013	Data completeness as a source of GIS uncertainty: commission and omission errors	Data Quality	Mixed quantitative and qualitative	Spatial feature inventory; data completeness assessment; commission and omission	Data availability	Commission rate and omission rate as standardised data completeness indicators	Commission rate; omission rate; completeness ratio per feature type	Data completeness (omission/commission) directly constrains the 'Data availability' Sobol factor; incomplete data reduces factor value and increases Sobol sensitivity (Figure 8; S1=0.40)	ISO 19157:2013, Section 4.4 (completeness); Sections 7.2.1-7.2.2 (commission and omission measures)	Data availability is the dominant Sobol factor (S1=0.40); completeness constraints are definitional
26	Chrisman (1982)	Positional error model for digitized cartographic features	Data Quality	Quantitative	Digitized cartographic vector data; positional accuracy of line and polygon features	Uncertainty	RMSE of digitized feature positions relative to source map coordinates	RMSE positional error; standard deviation of positional discrepancy	Establishes an early quantitative positional error model that informs the theoretical foundations of the 'Uncertainty' Sobol factor (Figure 8; S1=0.10)	Chrisman (1982), Proceedings AUTOCARTO 5: positional error characterisation for digitized cartographic data	Foundational positional error quantification; contributes to Uncertainty factor theoretical basis
27	Chrisman (1982)	Attribute classification error in cartographic data	Attributes	Quantitative	Cartographic attribute classification; categorical map accuracy	Uncertainty	Classification accuracy percentage for attribute categories in cartographic data	Classification accuracy; error matrix; per-class accuracy statistics	Establishes attribute classification error as a distinct GIS uncertainty source; contributes to the 'Uncertainty' Sobol factor and the Attributes error category in the pairwise interdependency matrix (Figure 8)	Chrisman (1982), Proceedings AUTOCARTO 5: attribute accuracy in cartographic data production and use	Early documentation of attribute classification error as a distinct GIS uncertainty type
28	Saltelli et al. (2008)	Sobol first-order index S1 computation: variance-based main-effect sensitivity	Modeling	Quantitative	Multi-parameter deterministic model; variance-based global sensitivity analysis	Weights	Sobol first-order sensitivity indices S1 for all four Sobol model factors	S1 = V(E[Y|Xi]) / V(Y); variance decomposition by individual input factor	Provides the S1 computation method applied in Figure 8 (S1: Data availability=0.40, Weights=0.30, Scale=0.20, Uncertainty=0.10);	Saltelli et al. (2008), Wiley: Chapter 4, variance-based sensitivity indices and S1 formula derivation	Primary methodological source for Sobol S1 indices in Figure 8; S1(Weights)=0.30

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									weights have the second-highest S1 index		
29	Saltelli et al. (2008)	Sobol total-effect index ST computation: variance including all interaction effects	Modeling	Quantitative	Multi-parameter model; total-effect sensitivity including higher-order interactions	Uncertainty	Sobol total-effect sensitivity indices ST for all four Sobol model factors	ST = 1 - V(E[Y|X~i]) / V(Y); total variance decomposition including interactions	Provides the ST computation method applied in Figure 8 (ST: Data availability=0.50, Weights=0.40, Scale=0.25, Uncertainty=0.15); ST-S1 differences quantify interaction effects	Saltelli et al. (2008), Wiley: Chapter 4, total-effect indices and interaction term quantification	Primary source for ST computation; ST minus S1 differences indicate parameter interaction effects
30	Herman and Usher (2017)	SALib Saltelli sampling for Sobol analysis: N*(2k+2) = 60,000 evaluations for k=4, N=10,000	Modeling	Quantitative	Python SALib library; Saltelli quasi-random sampling scheme; sensitivity analysis implementation	Uncertainty	Sobol S1 and ST with bootstrap confidence intervals from N=10,000 Saltelli samples	Saltelli sampler; N*(2k+2) model evaluations; SALib v1.4 implementation in Python 3.11	SALib v1.4 (SALib.sample.saltelli, SALib.analyze.sobol) was used to generate N=10,000 samples and 60,000 model evaluations for the Sobol analysis in Figure 8	Herman and Usher (2017), Journal of Open Source Software 2(9): 97; SALib documentation for Saltelli sampler	Direct software reference for SALib v1.4 and Saltelli sampling used in Figure 8
31	Monmonier (2018)	Choropleth classification scheme effects on map communication	Communication	Qualitative and illustrative	Thematic choropleth map; data classification; visual communication of spatial patterns	Weights	Perceived spatial pattern as a function of data classification scheme choice	Visual pattern assessment; class boundary sensitivity across classification methods	Classification scheme choice is a weighting decision that affects communication of GIS analysis results; maps the Communication error category and Weights factor in the pairwise interdependency matrix (Figure 8)	Monmonier (2018), How to Lie with Maps, 3rd ed., University of Chicago Press: Chapter 7 on classification schemes	Classification as a weighting decision links Communication error category to Weights Sobol factor
32	Monmonier (2018)	Scale bar and north arrow omission as communication errors in map output	Communication	Qualitative	Cartographic map output; map marginalia; communication standards compliance	Data availability	Presence or absence of essential cartographic metadata elements (scale bar, north arrow)	Binary presence/absence audit; cartographic standards compliance assessment	Communication data availability (map metadata completeness) contributes to the Data availability error category; absence of scale bar constitutes a scale communication error relevant to Figure 8	Monmonier (2018), How to Lie with Maps, 3rd ed., University of Chicago Press: Chapter 2 on scale and map design	Map metadata completeness is a data availability issue within the Communication error category

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33	Kedron et al. (2019)	Computational reproducibility in spatial analysis: assessment of reproducibility rates	Data Quality	Quantitative	Spatial analysis workflows; computational reproducibility; software and data availability	Data availability	Reproducibility rate (percentage of spatial analysis studies that are reproducible) as a data quality indicator	Reproducibility rate (%); binary reproducibility outcome per assessed study	Reproducibility failures are a form of data availability constraint; low reproducibility rates support the high Sobol sensitivity of the 'Data availability' factor (Figure 8; S1=0.40)	Kedron et al. (2019), Geographical Analysis 51(4): 382-390; Table 1 on reproducibility assessment outcomes	Reproducibility failures are data availability constraints; supports S1(Data availability)=0.40
34	Nust et al. (2018)	GIScience reproducibility: AGILE conference papers data and code availability evaluation	Data Quality	Quantitative	GIS research papers; data and code availability; open science practice in GIScience	Data availability	Percentage of AGILE conference papers providing publicly available data and code	Percentage papers with public data; percentage papers with public code	Provides empirical evidence for systematic data availability limitations in GIScience; low availability motivates the high Sobol sensitivity of 'Data availability' (Figure 8; S1=0.40, ST=0.50)	Nust et al. (2018), D-Lib Magazine 23(1/2): reproducibility assessment of AGILE papers	Empirical evidence for data availability constraints in GIScience; supports dominant S1(Data availability)=0.40
35	Bivand et al. (2013)	CRS transformation errors in R spatial objects: positional offset from misspecification	CRS/Datum	Quantitative	Projected coordinate systems; CRS transformation; R spatial data workflows	Data availability	Positional offset magnitude resulting from CRS misspecification in R spatial analysis	Positional offset (m); transformation accuracy assessment	CRS transformation errors reduce effective data availability by introducing positional offsets in DRASTIC input layers; contributes to the 'Data availability' Sobol factor (Figure 8; S1=0.40)	Bivand et al. (2013), Applied Spatial Data Analysis with R, Springer: Chapter 2 on CRS specification and transformation	CRS transformation error is a data availability concern for DRASTIC input layer preparation
36	Bivand et al. (2013)	Spatial weights matrix sensitivity in spatial regression modelling	Modeling	Quantitative	Spatial regression; spatial weights matrix; neighbourhood definition alternatives	Weights	Regression coefficient and likelihood ratio test statistic as functions of spatial weights specification	Regression coefficient variance; LR test statistic across weights matrix specifications	Spatial weights matrix sensitivity in spatial regression is directly analogous to DRASTIC weight parameter sensitivity; provides a modelling analogue for the 'Weights' Sobol factor (Figure 8; S1=0.30)	Bivand et al. (2013), Applied Spatial Data Analysis with R, Springer: Chapter 9 on spatial regression and weights	Spatial weights specification sensitivity underpins the Weights factor in the Sobol analysis
37	Longley et al. (2015)	Spatial data uncertainty principle in GIS: qualitative typology of uncertainty dimensions	Data Quality	Qualitative and illustrative	GIS spatial data; measurement uncertainty; data quality dimensions	Uncertainty	Qualitative typology of spatial data uncertainty dimensions and their sources	Qualitative uncertainty taxonomy; conceptual	Provides the conceptual framework for the 'Uncertainty' Sobol factor; maps uncertainty typology	Longley et al. (2015), Geographic Information Science and Systems, Wiley 4th	Conceptual basis for the Uncertainty Sobol factor (S1=0.10, ST=0.15) in Figure 8

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								uncertainty typology for GIS	to Figure 8 parameters and eight error categories in Case 10	ed.: Chapter 6 on uncertainty	
38	Longley et al. (2015)	Scale and resolution as dimensions of spatial data quality in GIS	Scale/Resolution	Qualitative and illustrative	Spatial data quality; scale; resolution; spatial data hierarchy	Scale	Qualitative framework for scale and resolution effects on GIS data quality	Qualitative scale taxonomy; resolution typology for raster and vector GIS data	Provides the conceptual basis for the 'Scale' Sobol factor; GIS scale dimensions are mapped to the Scale parameter tested in Figure 8 (S1=0.20, ST=0.25)	Longley et al. (2015), Geographic Information Science and Systems, Wiley 4th ed.: Chapter 3 on scale and resolution	Scale and resolution dimensions provide the conceptual underpinning for Scale factor in Figure 8
39	NRCan (2023)	National Road Network positional accuracy specification: +/-5 m CE90	Data Quality	Quantitative	National Road Network (NRN); road centrelines; authoritative positional accuracy specification	Data availability	CE90 positional accuracy standard for NRN road network features	CE90 (circular error at 90% probability); absolute positional accuracy specification	Defines the data quality threshold for authoritative road network data availability; illustrates the 'Data availability' Sobol factor constraint using a government-issued accuracy specification (Figure 8; S1=0.40)	NRCan (2023), National Road Network product specification: positional accuracy requirements section	Authoritative data quality specification illustrating Data availability factor constraints
40	Wilkinson et al. (2016)	FAIR data principles and GIS data interoperability: Findability, Accessibility, Interoperability, Reusability	Data Quality	Qualitative	GIS research data management; FAIR compliance; data interoperability standards	Data availability	FAIR compliance as a framework for data availability and interoperability in GIS research	FAIR compliance assessment; binary compliance per FAIR sub-principle	FAIR principles directly define the sub-dimensions of the 'Data availability' Sobol factor (Findability and Accessibility); FAIR non-compliance is a data availability constraint (Figure 8; S1=0.40, ST=0.50)	Wilkinson et al. (2016), Scientific Data 3: 160018; FAIR principles definition, rationale, and guiding principles	FAIR Findability and Accessibility map to Data availability (S1=0.40, the dominant Sobol factor)

Supplementary Table S2. Complete Sobol sensitivity analysis output: first-order (S1) and total-effect (ST) indices for four input factors of the DRASTIC groundwater vulnerability model (N = 10,000 Saltelli samples; 60,000 model evaluations). Presented in Figure 8.

Parameter	First-Order Index (S1)	Total-Effect Index (ST)
Data availability	0.40	0.50
Weights	0.30	0.40
Scale	0.20	0.25
Uncertainty	0.10	0.15

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